SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I A.Mohammed Kalifa (192411078) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

		and coherence			
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

1. Design File Transfer

Given:

- File size = **150 MB**
- Bandwidth = **50 Mbps**
- Segment size = **1500 bytes**
- Frame overhead = **40 bytes**

Calculation

Total segments

$$150 \times 1024 \times 1024 = 157{,}286{,}400 \text{ bytes}$$
$$\frac{157{,}286{,}400}{1500} = 104{,}858 \text{ segments}$$

Frames transmitted

= **104,858 frames**

Total Data Link overhead

$$104{,}858 \times 40 = 4{,}194{,}320 \text{ bytes}$$

≈ **4 MB**

Transmission time

150 MB = 1200 Mb

$$\frac{1200}{50} = 24 \text{ seconds}$$

Answer

- Segments = **104,858**
- Frames = **104,858**
- Overhead = **4,194,320 bytes**
- Transmission time = **24 seconds**

Explanation

- Transport Layer divides the file into segments and provides reliable delivery using acknowledgements and retransmission.
- Data Link Layer adds headers/trailers, detects errors using CRC, and delivers frames correctly over the physical link.

2. Sliding Window

Given

- Window size = 6
- Total frames = 48

Calculation

Windows required

$$48/6 = 8$$

One frame lost in each window
Retransmissions = **8**
Answer

- Windows required = **8**
- Retransmissions = **8 frames**

Flow Control
Flow control prevents the sender from sending data faster than the receiver can process, reducing congestion and improving efficiency.

3.IP Fragmentation

Given
Packet size = **12,000 bytes**
MTU = **1500 bytes**
IP Header = **20 bytes**
Calculation
Payload per fragment

$$1500 - 20 = 1480 \text{ bytes}$$

Fragments

$$12000/1480 = 8.1$$

Therefore,
Fragments = 9
Header overhead

$$9 \times 20 = 180 \text{ bytes}$$

Answer

- Fragments = **9**
- Header overhead = **180 bytes**

Explanation
The Network Layer fragments packets according to the MTU and reassembles them correctly at the destination using fragment offset and identification fields.

4. Sliding Window

Same as Question 2.
Answer

- Windows = **8**
- Retransmissions = **8**

Flow control improves communication by avoiding buffer overflow and maintaining efficient data transfer.

5. Session Layer

Given
Video = **600 MB**
Checkpoint every **60 MB**
Failure after **390 MB**
Calculation
Checkpoints before failure
60,120,180,240,300,360
= **6 checkpoints**
Last checkpoint = **360 MB**
Retransmission

$$390 - 360 = 30MB$$

Answer

- Checkpoints = **6**
- Retransmitted data = **30 MB**

Explanation
Synchronization checkpoints allow transmission to resume from the last checkpoint instead of restarting from the beginning.

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6. Presentation Layer

Given
Original file = **900 MB**
Compression = **40%**
Encryption overhead = **5%**
Calculation
Compressed size

$$900 \times 0.60 = 540MB$$

Encrypted size

$$540 \times 1.05 = 567MB$$

Reduction

$$900 - 567 = 333MB$$

Percentage reduction

$$\frac{333}{900} \times 100 = 37\%$$

Answer

- Compressed size = **540 MB**
- Encrypted size = **567 MB**
- Reduction = **37%**

Explanation
Presentation Layer compresses data to reduce transmission size and encrypts data to improve security.

7. FTP Transfer

Given
Data = **3 GB**
Throughput = **120 Mbps**
Request size = **3 MB**
Calculation
3 GB = 3072 MB
Requests

$$3072/3 = 1024$$

Transmission time
3 GB = 24,576 Mb

$$24576/120 = 204.8 \text{ s}$$

Answer

- Requests = **1024**
- Transmission time = **204.8 seconds**

Explanation
The Application Layer provides services like FTP for reliable file transfer, user authentication, and error handling.

8. End-to-End Delay

Processing delays
Application = 4 ms
Transport = 3 ms
Network = 5 ms
Data Link = 2 ms
Physical = 1 ms
Propagation = 18 ms
Transmission = 12 ms
Calculation

$$4 + 3 + 5 + 2 + 1 + 18 + 12 = 45ms$$

Answer
Total end-to-end delay = **45 ms**
Explanation
Each OSI layer performs specific functions such as data formatting, routing, framing, and transmission, contributing to the total delay.

9. Frame Overhead

Given
Useful data = **80 MB**
Frame size = **1600 bytes**
Overhead = **80 bytes**
Payload

$$1600 - 80 = 1520 \text{ bytes}$$

Useful data

$$80 \times 1024 \times 1024 = 83{,}886{,}080 \text{ bytes}$$

Frames

$$83886080/1520 = 55189$$

Overhead

$$55189 \times 80 = 4{,}415{,}120 \text{ bytes}$$

Efficiency

$$\frac{1520}{1600} \times 100 = 95\%$$

Answer
- Frames = **55,189**
- Total overhead = **4,415,120 bytes**
- Efficiency = **95%**

Explanation
Protocol overhead consumes bandwidth, reducing transmission efficiency, although it is necessary for addressing, error detection, and reliable communication.

10. Hospital IoT Network

Given
Original file = **240 MB**
Compression = **30%**
Segment size = **1400 bytes**
Frame overhead = **60 bytes**
Calculation
Compressed file

$$240 \times 0.70 = 168MB$$

Compressed bytes

$$168 \times 1024 \times 1024 = 176{,}160{,}768$$

Segments

$$176160768/1400 = 125830$$

Overhead

$$125830 \times 60 = 7{,}549{,}800 \text{ bytes}$$

Final transmitted data

$$176160768 + 7549800 = 183710568 \text{ bytes}$$

≈ **175.2 MB**

Answer

- Compressed file = **168 MB**
- Segments = **125,830**
- Protocol overhead = **7,549,800 bytes**
- Final transmitted data = **183,710,568 bytes (≈175.2 MB)**